



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PCM PETROLEUM AGENCIES IOC DE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

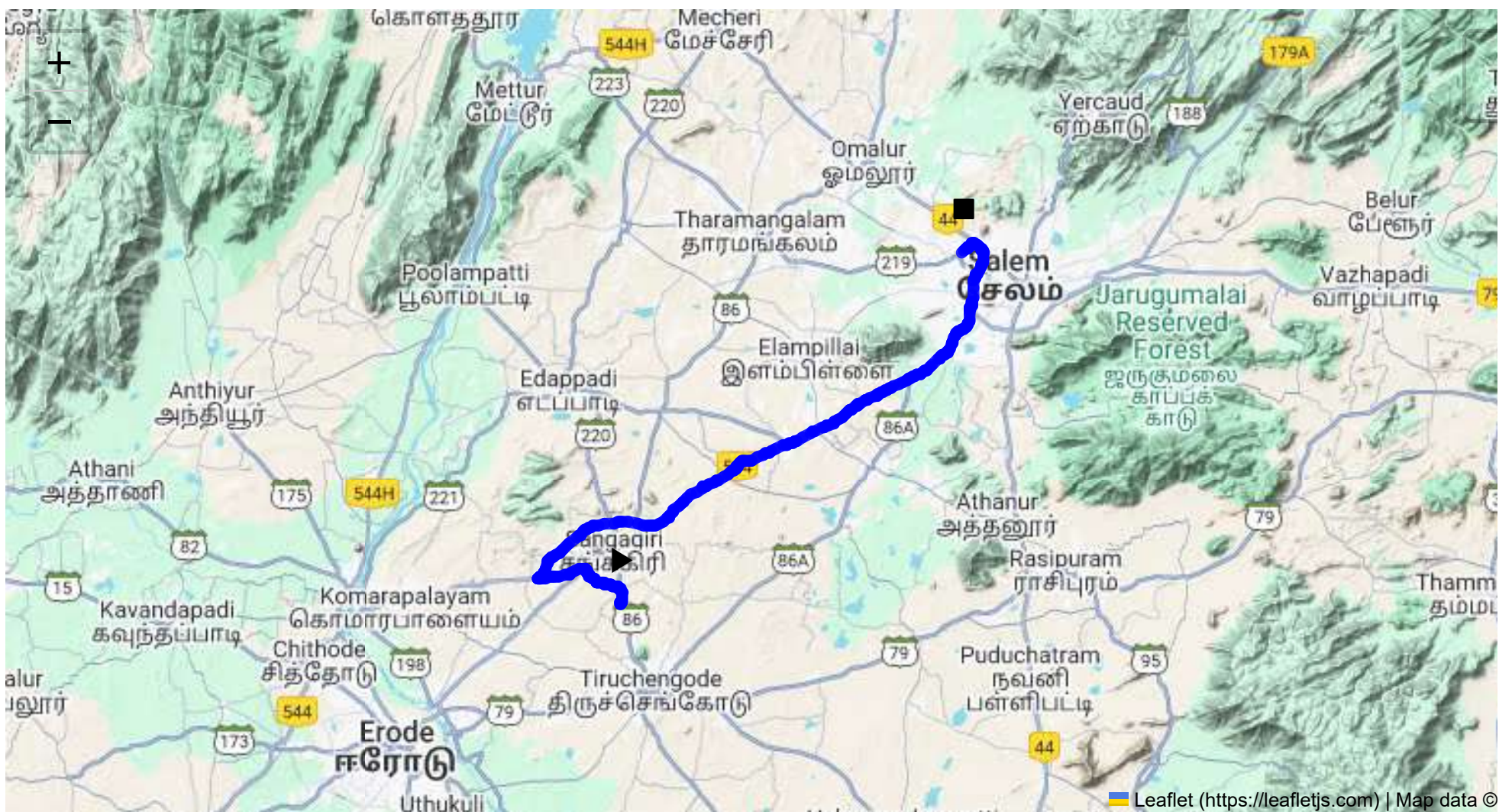
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 56.33 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.679132, 78.112572)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

This route begins at Sangagiri in Tamil Nadu and proceeds through Sankari RS, Sankari West, and Padaiveedu before reaching Salem. The route covers approximately 56.33 kilometers and primarily

follows major regional roads known for accommodating both local and through traffic.

2. Weather Conditions

- **Typical Weather:** The area experiences tropical weather, with hot summers, moderate winters, and a monsoon season between June and September.
- **Weather-related Hazards:** Monsoon rains can cause flooding, reducing visibility and increasing the risk of road disruptions. Summer heat may lead to surface tarmac issues.

3. Traffic Patterns

- **Peak Hours:** Traffic typically peaks during morning hours (8:00 AM to 10:00 AM) and evening hours (5:00 PM to 7:00 PM), especially near urban centers like Salem.
- **Congestion-Prone Areas:** Congestion is likely in market areas and urban gateways, particularly in Sangagiri and Salem. Intersections on Bhavani Main Road may also see congestion.

4. Road Quality and Infrastructure

Roads along the route vary from well-maintained highways to narrower roads with occasional maintenance needs. Potholes and uneven surfaces may be present, especially after monsoon rains.

5. Alternative Routes for Emergencies

In emergencies, consider:

- Using the NH44 (also known as the old NH7) for bypassing congested zones.
- Local knowledge-based diversions through parallel minor roads can be explored if major roads are blocked.

6. Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is subject to strict regulations requiring appropriate labeling, documentation, and adherence to safety protocols. Night travel may be restricted for such loads on some segments.

7. Historical Incidents

While there is no comprehensive database available publicly, local news archives and reports indicate accidents involving heavy vehicles on Bhavani Main Road, often linked to speeding or poor visibility.

8. Environmental Considerations and Sensitive Areas

- Some route segments are near agricultural land, necessitating cautious driving to avoid contamination.
- Industrial zones (e.g., near Salem) demand adherence to specific environmental protocols to prevent hazardous spills.

9. Communication Coverage

Mobile network coverage is generally adequate, but there may be intermittent dead zones in more rural stretches, particularly between Sankari and Padaiveedu.

10. Estimated Emergency Response Times

- Urban areas like Sangagiri and Salem: 20-30 minutes.
- Rural stretches and between smaller towns: 30-45 minutes, dependent on road conditions.

11. Summary of Risk Assessment

This route presents typical challenges of rural and semi-urban Indian roads: variable traffic conditions, weather influence, and infrastructure quality issues. Significant care is needed when driving through congestion-prone areas or during inclement weather. Emergency preparedness is crucial, given the potential for communication gaps and delayed response times. Proper documentation and adherence to regulations are mandatory for the transport of hazardous materials to ensure safety and compliance.

By preparing for these challenges, truck drivers can safely navigate from Sangagiri to Salem while carrying hazardous materials. Regular updates on road and weather conditions, along with emergency protocols, should be provided to transport personnel regularly.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	Medium	11.45352, 77.85698	30 KM/Hr
5	Turn	Medium	11.46048, 77.85036	30 KM/Hr
6	Turn	Medium	11.46303, 77.84945	30 KM/Hr
7	Turn	Medium	11.46318, 77.84913	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45631, 77.81668	15 KM/Hr
10	Turn	Medium	11.61748, 78.10604	30 KM/Hr
11	Turn	High	11.67074, 78.12668	15 KM/Hr
12	Turn	High	11.67081, 78.12661	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	High	11.68618, 78.12088	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
15	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
19	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
21	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
16	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
20	school	Mothers Vidthyalaya Metric School	11.6884655, 78.1240569	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61748, 78.10604



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67074, 78.12668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67081, 78.12661



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68618, 78.12088
